

Dilawar Singh

Senior Backend Engineer | SaaS, APIs & Platform Engineering

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Founding / Senior Backend Engineer with 10+ years of experience building high-performance APIs, SaaS platforms, and data pipelines in Rust, Python, and C++. Proven record of taking products from PoC to production: designing scalable microservice architectures, operating self-hosted infrastructure, and shipping reliable cross-platform systems. Enthusiastic contributor to Free and Open Source Software (FOSS).

Core Competencies: API Design (REST/gRPC) | Rust | Python | PostgreSQL | Microservices | SaaS Platforms | Observability | CI/CD | Team Leadership

Skills

- Languages: Rust, Python, C, C++ (expert); Java, TypeScript, JavaScript, PHP (intermediate)
- SaaS Platforms & APIs: High-performance gRPC and REST services using Axum, Actix, FastAPI, CodeIgniter, Laravel; API design and versioning; authentication and authorization
- Data: SQL, PostgreSQL, DuckDB; job queues; data parsing, validation, and lifecycle management; real-time event pipelines (~10,000 events/session)
- Observability: OpenTelemetry, Grafana; structured logging; incident response tooling
- Infrastructure & DevOps: Docker, QEMU; CI/CD (GitLab, Jenkins); build systems; packaging (OBS); Ansible, Pulumi; self-hosted infrastructure management
- Systems: Linux Kernel Modules, eBPF, Windows WDK, Win32 and POSIX APIs, IPC, SDK development (Rust, C++)
- Other: Parser development, event-driven systems, technical literature review

Work Experience

Independent Consultant — Self-Employed, Bengaluru

Mar 2026 – Present

- Designing and overseeing development of high-performance APIs (Rust, AMQP) for a cyber lab running near-real-time experiments on highly correlated, diverse datasets (audio, video, sensor events)
- Designing and developing scalable microservices supporting a virtual assistant platform for an automotive control system (Rust, C, Qualcomm SoC)

Senior Engineer — Dognosis, Bengaluru

Jan 2025 – Feb 2026

- Aggregated ~30 GB/session and ~10,000 events/session of live sensor data into a single operational control plane used for real-time decision-making and incident response
- Set up and operated self-hosted infrastructure (GitLab, PostgreSQL, Grafana, CI/CD runners), including access control, backups, and deployment policies
- Established risk-focused planning and sprint rituals that reduced invisible work, clarified ownership, and gave leadership visibility into team throughput

- Built hiring pipelines and hired 5 engineers, scaling the team from 6 to 11 in 3 months

Cofounder & CTO — Subconscious Compute, Bengaluru

Dec 2019 – Dec 2024

- Hired and led a team of 15 engineers to build Shepherd, an automated endpoint compliance and security SaaS platform
- Designed the core backend architecture (Rust/C++) for a cross-platform threat detection and mitigation engine; drove the platform from PoC to production across customer endpoints
- Authored technical memos and pitch decks to secure accelerator placements and support fundraising

Research Fellow & Google Summer of Code (GSoC) Mentor

2016 – 2018

- Wrote Python bindings (pybind11) and created packages for MOOSE and Smoldyn, significantly improving usability and adoption by the scientific community
- Added high-performance numerical solvers (C++, Boost, GNU Scientific Library) to the MOOSE simulator
- Mentored multiple Google Summer of Code (GSoC) contributors on GPU/CUDA-based acceleration over 4 years

Firmware Engineer — Kritikal Solutions, Noida

Jul 2009 – Jun 2010

- Implemented a Kalman-filter-based image stabilization system for professional movie cameras (C++, multi-threaded RTOS)

Projects

- Built an automated scheduling system for NCBS, eliminating thousands of manual emails and calendar conflicts (PHP, Vue.js)
- Published Rust crates for stream processing, online statistics, WMI interfaces, and CLI tooling
- Developed Python tools for scientific data extraction, instrumentation UIs, and internal developer tooling

Education

- PhD, Computational Neuroscience — NCBS Bangalore (TIFR Mumbai), 2019
- Doctoral Studies, Digital Systems — IIT Bombay, 2011–2013
- M.Tech, Electrical Engineering (VLSI) — IIT Bombay, 2009
- B.Tech, Instrumentation & Control Engineering — Dr. MGR ERI Chennai, 2007

Publications

[Full list on Google Scholar](#)

- [Python interfaces for the Smoldyn simulator](#), Bioinformatics, 2022
- [BioSimulators: a central registry of simulation engines](#), Nucleic Acids Research (NAR), 2022